

Label Forwarding (LFB): Built from LFB; used for actual forwarding decisions. (LFB only contains currently best LSP (decision: Routing Protocol) **Control Plane:** Builds routing/label tables (RIB, LFB) **Data Plane:** Forwards packets (FIB, LFB); pushes/swaps/pops labels

MPLS HEADERS

4-byte header before IP: With MPLS VPN, there will be a stack of two headers **Label (20b):** actual MPLS label **EXP (3b):** QoS/CoS, Now called Traffic Class (TC) **S-bit (1b):** Bottom of the label stack indicator, 1 = True = last label before IP header **TTL (8b):** time-to-live (equal to IP TTL)

TTL

- Ingress PE decrements IP TTL field & copies IP TTL field into new MPLS TTL
- routers decrements MPLS TTL
- Egress PE decrements MPLS TTL, pops final MPLS header, copies IP TTL
- If provider doesn't want to expose MPLS network: disable **MPLS TTL propagation** on PE, PE sets MPLS TTL=255, egress PE leaves original IP TTL unchanged → network appears as single router hop from TTL perspective

LABEL DISTRIBUTION PROTOCOL (LDP) - CONTROL PLANE

- Distributes labels to neighbors, triggered by new IP route in FIB
- **Hello messages** sent via UDP (Port 646) to 224.0.0.2 to discover neighbors
- TCP (646) connection used to exchange label bindings (prefix to local lbi)
- Routers advertise all local bindings after TCP session is up
- Label mapping used to build LFB → LFB
- LDP router ID must be reachable (via routing table)
- Each router manages local labels independently
- Relies on underlying routing protocol for best route & loop prevention

MPLS L3 DATA PLANE

Outer label: Transport label (LDP); identifies LSP between ingress/egress PE **Inner label:** VPN label (MP-BGP); identifies customer VRF **Push:** Ingress PE; classify and label packets **Swap:** P router; replaces label, forwards based on new label **Pop:** Egress PE removes label; sends original packet to CE **Penultimate Hop Popping (PHP):** penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

VIRTUAL ROUTING AND FORWARDING (VRF) TABLES

- Virtual router inside a PE. Maintains isolated RIB + FIB and separate routing process per CE (VPN isolation)
- Exists per MPLS-aware PE router; one per attached customer

ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)

Route Distinguisher (RD): Uniquely identifies VPN routes – allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN Used to **make routes unique** in BGP. Forms **VPNs V4/NLRI (MP-BGP): RD:IPv4 prefix** **Route Target (RT):** Controls route **import/export** between VRFs. Used as **extended BGP community path attributes**. Typically in the form **ASN:nn** or **IP:nn**, e.g., 65000:100, 1:10

How VRFs Work: Route is tagged with an RT when advertised by BGP. Other VRFs import the route if the RT matches their import policy. Allows overlapping or shared connectivity between tenants (e.g., shared services).

OVERLAY TECHNOLOGIES

Overlay network (virtual) sits on top of the **underlay network** (physical). **Generic Routing Encapsulation (GRE):** Encapsulates data packets. Unencrypted, point-to-point Enables usage of not supported protocols. Carries a passenger protocol (the original packet) inside a carrier/transport IP header that the underlay routes

SEGMENT ROUTING (SR)

Source routing: Sender defines full path using Segment (= Instruction) List **Segment Identifier (SID):** Each SID = 1 instruction (e.g., forward via ECMP, specific face, or to a service) **Segment List:** Ordered SID list carried in packet header; defines full route **State in packet:** No per-flow state, intermediate nodes follow SID instructions **No to new control plane:** Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed

Data plane: Can operate over MPLS or IPv6 **Simple but powerful:** Enables TE, fast reroute, policy routing

SEGMENT LIST OPERATIONS

Push: Insert SIDs into packet; set active SID (top of list) **Continue:** Active SID not yet completed; keep processing it **Next:** Current SID completed; activate next SID in list

SEGMENT IDENTIFICATION

Global Segments: Known and supported by all SR nodes in the domain, installed in forwarding tables across the network (e.g. "Forward packet according to shortest path to Node1"). Defined in the SR Global Block (SRGB) and should be consistent across all nodes

Local Segments: Defined and installed only on originating node, Not forwarded by others, but must be understood network-wide (e.g. "Forward packet on interface to Node2"). Defined in the SR Local Block (SRLB) and are specific to the local SR node.

CONTROL PLANE SEGMENT TYPES

IGP Prefix Segment: Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries, "steer traffic along ECMP-aware shortest path" **IGP Node Segment:** Global SID for a specific node (shortest-path forwarding) **IGP Anycast Segment:** Global SID for a group of nodes; traffic sent to nearest IGP Adjacency Segment: Local SID; direct link to neighbor **L2 Adjacency Segment:** Local SID for Layer-2 segment (e.g., Ethernet link) **Combining Segments:** End-to-end paths can mix IGP and BGP segments. Traffic to BGP Anycast → more ECMP in data centers

SR-MPLS

- Reuses existing MPLS data plane – no hardware change needed
- Segments = MPLS labels; Segment List = label stack (top = active)
- Segments distributed via IGP/BGP; no LDP necessary (but possible)
- Supports both IPv4 and IPv6 networks
- PUSH= PUSH, CONTINUE= SWAP, NEXT= POP

BENEFITS OF SEGMENT ROUTING

Benefits: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust, Network Innovation **Source Routing:** Balances distributed intel with centralized optimization **Traffic Engineering (TE):** Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS

DRAWBACKS OF TRADITIONAL NETWORKS

Control Plane: LDP/RSVP-TE adds complexity **Scalability:** Per-flow/path state limits growth; LSP and signaling overhead increase rapidly

VIRTUAL EXTENSIBLE LOCAL AREA NETWORK (VXLAN)

Issues of L2: STP, Max amount of VLANs (4094), Large MAC Address tables **VXLAN:** Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation

VXLAN Network Identifier (VNID): 24-bit identifier (up to 16 million segments) that defines the VXLAN broadcast domain.

Virtual Tunnel Endpoint (VTEP): Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.

Network Virtual Interface (NVI): Logical interface on a VTEP used for VXLAN tunnel operations.

FRANE FORMAT

- Original Ethernet frame → VXLAN Header → UDP → Outer IP Header
- The VXLAN header contains the 24-bit VNID and flags.
- Outer headers allow Layer 2 frames to traverse IP underlay networks.

VIRTUAL NETWORK IDENTIFIER (VNI)

- 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
- Replaces traditional VLAN IDs (12-bit)
- Used by VTEPs to map traffic into corresponding Layer 2 domains.

VXLAN TUNNEL ENDPOINT (VTEP)

Connects the overlay (VXLAN) and underlay (IP) networks. AVTEP can have multiple VNI interfaces, but they associate with the same VTEP IP interface. **Software VTEP:** Located on hypervisors using virtual switches. **Hardware VTEP:** Located on routers/switches with ASICs for performance. **VTEP IP Interface:** Connects to the underlay network, handles encapsulation. **VNI Interface:** Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

MAC ADDRESS LEARNING

On control plane: happens proactively **On data plane:** ad-hoc with flooding

- Each VTEP maintains a **VXLAN mapping table** linking destination MAC addresses to remote VTEP IPs.
- **Learning via ARP:**
 - Host H1 sends ARP request, switches learn H1's MAC.
 - ARP request is flooded to H2.
 - H2 responds; switches learn H2's MAC.
- **Learning Methods:**
 - BUM traffic is inefficient.
 - **Multicast VXLAN:** VTEPs join multicast groups per VNI. Scales better, offloads BUM replication. 20+ VTEPs = there is too much traffic, doesn't scale well
 - **MP-BGP EVPN:** Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

ETHERNET VIRTUAL PRIVATE NETWORK (EVPN)

Overcome flood-and-learn limitations, doesn't rely on data plane learning, utilizes robust control plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, L2 and L3 Support.

LAYER 2

L2 bridging across L3 networks **BGP Control Plane:** Distributes MAC info (no flooding → efficiency) **VXLAN Overlay:** Encapsulates L2 in L3 UDP (data plane)

Multi-Tenancy: via VNI segmentation **Redundancy:** All-active multihoming, ECMP, fast convergence

USE CASES

- Multi-tenant datacenter interconnects (DCI)
- Extending L2 over WAN between remote sites
- Scalable, segmented L2 fabrics

AUTODISCOVERY VIA ROUTE REFLECTORS

- RR reflects VNI routes to other PEs
- RR doesn't participate in EVPN or pseudowires
- RR needs only **address-family l2vpn evpn**
- L2VPN RIB stores endpoint/VFI info for control plane
- **BGP UPDATE** from spines contain **ORIGINATOR_ID** (origin leaf)

HOST DETECTION

- Host connects to VTEP → MAC learned locally
- VTEP advertises MAC + L2VNI via BGP EVPN
- MAC learning follows normal Ethernet semantics

INGRESS REPLICATION (IR)

BUM traffic, when Multicast underlay network is not used, handle multi-destination traffic (ARP → unicast)

EARLY ARP TERMINATION (ARP SUPPRESSION)

- Avoids flooding ARP requests
- VTEP queries control plane for MAC/IP/VNI mapping
- If known → direct unicast (no broadcast)

Silent Host Flow (Fallback)

- If IP/MAC unknown → ARP sent via **ingress replication**
- Replicated ARP request goes to remote VTEPs
- Only correct host responds → update reflected to all VTEPs
- Future traffic uses updated BGP mapping

LAYER 3

By adding L3 features, data routing efficiency + smooth connections improve

EVPN ROUTE TYPES

(1) Ethernet Auto-Discovery Route (3) Inclusive Multicast Route (4) Ethernet Segment Route (6) Selective Multicast Ethernet Tag Route (7) IGMP Join Synchrony Route (8) IGMP Leave Synchrony Route **Type 2 – Host Advertisement:** Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when host connects to VTEP. **Type 5 – Subnet Advertisement:** Advertises IP prefix + prefix length with L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.

HOST DETECTION

- Host sends ARP/ND to local VTEP
- VTEP learns MAC/L2VNI and IPL3VNI
- Info is advertised in EVPN (control plane)

HOST DELETION & MOVE

Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IPL3VNI advertisements.

Host Move: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

VIRTUAL ROUTING AND FORWARDING (VRF)

- Supports independent policies per tenant
- Key for scaling and multi-customer separation

DISTRIBUTED ANYCAST GATEWAY (DAG)

- Same gateway IP+MAC on all VTEPs
- Enables local default gateway for hosts in MPLS network with IRB
- Supports mobility + optimal forwarding across BGP EVPN fabric

INTEGRATED ROUTING AND BRIDGING (IRB)

- Enables inter-VLAN routing inside EVPN
- Avoids central gateway → no "traffic tromboning"

SYMMETRIC IRB (L2 + L3)

- Routing/bridging on ingress + egress VTEPs
- Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
- Scales well; clean separation of MAC and IP
- Requires L3 connectivity between all source/destination VTEPs for Type 2 routing (Done by configuring Type 5 routing)

ASYMMETRIC IRB (L2)

- Routing only on ingress, bridging on egress
- VXLAN uses destination VNI in both directions
- One L2 VNI per VLAN/Subnet
- Simple config, but requires all VLANs/VNIs on all VTEPs (scaling issues)

MULTIPROTOCOL BGP (MP-BGP)

Address Family Identifier (AFI): Category of information being carried **Subsequent AFI (SAFI):** Narrows down the specific type within that category **EVPN NLRI:** Uses MP-BGP with specific AFI/SAFI. Supports multiple route types and attributes, unsupported routes are dropped by BGP

- Enables protocol-based VTEP discovery and host reachability via control-plane learning
- Reduces flooding by replacing data-plane learning
- Extends BGP with multiprotocol capabilities (AFI/SAFI)
- Uses **MP_REACH_NLRI** and **MP_UNREACH_NLRI** for route advertisement and withdrawal
- Uses the **VPV4v4** NLRI to distinguish between duplicate IPv4 prefixes

BGP CONTROL PLANE

- PEs learn MACs from local CEs (data plane)
- MACs advertised via BGP (control plane)
- Uses Route Distinguishers and MPLS labels
- Remote PEs update L2 RIB/FIB with MAC and next-hop info
- Enables seamless L2 across IP/MPLS backbone

CDN

Origin Server: Central content source (original files), usually in a datacenter **Edge / CDN Server:** Geographically distributed, caches content. Also called **Point of Presence (POP)**

DNS Infrastructure: Directs users to optimal edge server (e.g. Geo-Routing)

KEY BENEFITS

Latency Reduction: Nearby edge servers reduce round-trip time **Availability:** Failover and redundancy in case of node failure **Scalability:** Handles traffic spikes via load balancing **Cost Optimization:** Reduces backend and transit load on origin **DDoS Protection:** Edge servers absorb attacks → not all traffic on one server **Global Load Reduction:** Less long-distance traffic across the Internet

REQUEST ROUTING TECHNIQUES

- Decides which edge server should serve a client request
- Goal: Best performance (e.g. proximity, load, responsiveness)

DNS-BASED GEO-ROUTING

- Each edge has a unique IP
- DNS server picks closest/optimal edge server based on:
 - Resolver IP location (not user! → can cause wrong choice)
 - GeoIP DBs (MaxMind, IP2Location), load, latency, business rules

EDNS(0) and Client Subnet Extension (ECS):

- Resolver includes part of client IP in DNS request (e.g. /24 subnet)
- Authoritative DNS makes better decision based on actual client region
- Improves accuracy without revealing full IP

ANYCAST WITH BGP

- L2VPN RIB stores endpoint/VFI info for control plane
- BGP routing decides which path is "best" (AS-path, local pref, etc.)
- No DNS logic or per-client decision – pure BGP convergence

Pros: Fast failover, simple, no app logic needed **Cons:** Less control, BGP ≠ best latency, route flapping risk

HTTP CACHING & HEADERS

Caching is controlled via HTTP headers between clients, proxies, and servers **Cache-Control:** Main directive (**no-cache, no-store, max-age, must-revalidate**, etc.)

Expires: Absolute expiration time (older method, replaced by **Cache-Control**) **Etag:** Validator tag (version/hash), used with **If-None-Match**

Last-Modified: Timestamp used with **If-Modified-Since** for revalidation **Age:** Time (in seconds) since response was fetched from origin **Validation:** Client uses **Etag** or **Last-Modified**; server → 304 if unchanged

QOS

Internet is best effort: No guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization **Quality of Experience (QoE):** Perceived service quality from user perspective **Route Pinning:** Keeps flow on a fixed path to prevent oscillation (don't switch immediately to "better" path)

NETWORK PERFORMANCE METRICS

Latency / Delay [ms]: time for packets to travel src → dest (Voip<150ms) **End-to-End Delay:** Total time sender to receiver **One-Way Delay:** From first bit sent to last bit received **Delay Components:** **Transmission delay** (time to push onto link), **Processing delay** (lookup, queuing), **Propagation delay** (physical travel time) **Jitter [ms]:** Variation in delay between packets, caused by re-routing/queuing (Voip<30ms). Calc no queue, queue delay. Doesn't impact TCP **Throughput:** Rate of successfully delivered data **Packet Loss [%]:** Dropped packets due to congestion or errors (Voip<1%) **Bandwidth [Gbit/s]:** Maximum transfer capacity of a link

QUEUEING ALGORITHMS

First-In First-Out (FIFO): Basic, no prioritization **Priority Queueing (PQ):** Multiple queues, serve highest first; others may starve **Round-Robin (RR):** One queue per class, serve in turn (fair, but ignores priority) **Weighted Fair Queueing (WFQ):** RR with weights, n packets per queue **Class-Based WFQ (CBWFQ):** WFQ with user-defined classes, queue limits, max bandwidth guaranteed or max % of bandwidth (logical queues based on IP Precedence only)

Low Latency Queueing (LLQ): Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol...)

QUEUE MANAGEMENT

Tail Drop: Drops packets when queue full; huge interruption of traffic → same as no connectivity **STP Global Sync:** Many TCP flows back off and restart simultaneously → link underutilization

TCP Starvation: TCP slows down after drops, UDP doesn't → queues filled with UDP, TCP squeezed out

Random Early Detection (RED): Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their windows sizes

Weighted RED (WRED): RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic

TODD:

DSCP / EXP: DSCP (6-bit in IP header) marks packets for QoS; used in DiffServ for classifying traffic. EXP (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP.

POLICING VS. SHAPING

Policing (Inbound outburst): Drops packets that exceed configured rate limits **Shaping (Outbound):** Buffers packets to smooth traffic bursts

QOS MODELS

Best Effort: No guarantees, all traffic treated equally (follow Internet neutrality) **Integrated Services (IntServ):** End-to-end QoS, per-flow resource reservation, precise but not scalable (uses RSVP) **Differentiated Services (DiffServ):** Class-based, scalable approach using marking, no hard guarantees

TRAFFIC MARKING

L3 Marking: 10s byte → DSCP (6 bits) + IP Precedence (3 bits) **L2 Marking:** Dot1q header → 802.1p CoS bits

MODULAR QOS CLI (MQC)

Class Map: Define traffic classes (e.g., match voice or video) **Policy Map:** Define actions for each class (e.g., limit, shape, priority) **Service Policy:** Apply policies to interfaces or directions (in/out)

OTHER INFOS

AD: Inter-protocol choice (e.g., OSPF vs RIP) → lower wins. **Cost/Metric:** Intra-protocol choice (e.g., OSPF path A vs B) → lower wins. **Routing Preference Order (across protocols):**

- Most specific prefix
- Lowest Administrative Distance
- Static default route

Administrative Distances: (Smallest Administrative Distance wins)

	Protocol	Distance
Connected		0
Static (Interface)		1
Static (Next Hop)		1
BGP External		20
EIGRP Internal		90
OSPF		110
ISIS		115
RIP v1/v2		120
EIGRP External		170
BGP Internal		200

EVPN BGP ROUTING TABLE INFOS

= Would not be there if it was L2 VNI BGP Routing Table **Route Distinguisher:** 172.16.255.101:32777 **Route Type:** 2 **MAC Address Length:** 48 **MAC Address:** 5254.00f8.29a8 **IP Address Length:** 32 **IP Address:** 10.10.0.100 **L2 VNI:** 30010 **L3 VNI:** 50000 **Remote VTEP IP Address:** 172.16.254.101 **L2 Route Target:** 1:10 **L3 Route Target:** 65000:50000

Leaf-03# show bgp l2vpn evpn 10.10.0.100 **BGP routing table information for VRF default, address family L2VPN EVPN** **Route Distinguisher:** 172.16.255.101:32777 **BGP routing table entry for** [2]:[0]:[0]:[48]:[5254.00f8.29a8]:[32]:[10.10.0.100]/272, version 19897 **Paths:** (1 available, best #1) **Flags:** (0x000202) (hlg32 00000000) on xmit-list, is not in l2r1b/evpn, is not in HW **Advertised path-id 1** **Path type:** internal, path is valid, is best path, no labeled nexthop **Imported to 2 destination(s)** **AS-Path:** NONE, path sourced internal to AS 172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1) **Origin:** IGP, MED not set, localpref 100, weight 0 **Received label:** 30010 50000 **Extcommunity:** RT:1:10 RT:65000:50000 ENCAP:8 Router MAC:5254.00ca.69ae **Originator:** 172.16.255.101 **Cluster List:** 172.16.255.1

TODD:

```
<begin{center}>
*Route Type 2:*
<includegraphics[width=\linewidth]{route-type-2}>
*Route Type 5:*
<includegraphics[width=\linewidth]{route-type-5}>
<end{center}>
```

PRUFUNG VORJAHR

NETWORK DESIGN

3-tier campus network: Default Gateway (D), QoS marking (A), STP Root Port (A), HSRP, VRRP or GLBP (D), "Simple" (C), OSPF Totally Stub Area (D), High availability (C)

Campus Design: used to reduce size of L2 domain: EVPN, MPLS

REST

MP_REACH_NLRI: Next hop, MAC Address

- VXLAN is a data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network.
- The underlay network is unaware of VXLAN devices that connect to the physical switches are unaware of VXLAN.
- A route distinguisher is used to uniquely identify a route in combination with the destination prefix.

TODO'S
- p.74,75 - OSPF + IS-IS path cost calculation + path choosing - OSPF has to go through backbone
OSPF
- passive interfaces - flooding - O N1 / O N2 routes - route summarization - synchronizing the lsdb - Fast Reroute (FRR)
IS-IS
- ES-IS, IS-IS: details
BGP
- IBGP split horizon - comparison to IGP - Multihop sessions - NLRI - Aggregate impact (TE) - rework - Best path calculation - Idle → Connect → Active → OpenSent → OpenConfirm → Established - MED explanation - default external cost = 20?
AVAILABILITY
- everything: thanks, past me, that was very informative
MULTICAST
- Link-local multicast (48) - IPv4 MAC Address mapping - IGMP and the querier/Role in Dense Mode/Challenges
TOPOLOGY
- p.59-61 - Slides 40-50 - pros/cons of different topologies
WAN
EVPN
- Route types - IRB diagram
QoS
- Types of traffic - Queuing input/output buffer - IntServ vs DiffServ - QoS classification - DSCP:PHB,CS,AF,EF - rest - Node based vs Routing based?
TODO:
- github repos for IS-IS and OSPF labs - rendezvous point nochmal anschauen - network design components - classic, old, enterprise, evpn etc - next-hop-self - why flagp needed for ebgp - ip address otherwise might not be known in AS internally - cdn lab (tags?) - qos: why do we need it? - how does it work conceptually - voice QoS sensitive to? - spick lukas QoS+
TODO:
notes 37+